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Economic recession and the nature and pace of social change in Japan

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ABSTRACT

Previous research has found a link between economic development and post-materialist and libertarian values. This research examines social change in values in Japan from the mid-1970s to the beginning of the 21st century. Using Japanese survey data, we investigate the nature and pace of social change during the economic crisis of the early 1990s, as well as in the periods before and after the burst of the bubble economy. We find that social change stalled during the economic crisis and resumed again in the late 1990s, but at a much lower level than in the late 1980s. We argue that economic period effects are largely responsible for the slowdown in social change in values in Japan at the end of the 20th century.

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1. Introduction

What happens to social change during periods of economic turmoil? Does the progress towards libertarian and post-materialist values that has been linked to economic development continue? Or do individuals focus on basic economic needs during periods of economic hard times—thinking little about higher-order needs and post-materialist ideals in the face of increased job insecurity and uncertainty? Do some even retreat to more traditional values out of nostalgia for better economic days?

This paper sets out to examine the question of social change in values during a period of economic hard times in Japan following the burst of the asset price bubble in the early 1990s. Japan provides a particularly suitable case for analysis because rapid modernization and economic development during the mid 20th century was followed by a severe economic crisis when soaring land prices crashed in the early 1990s. We set out to examine the nature and pace of social change in Japan during this tumultuous period in the country's economic history. The analysis addresses general questions about cultural change and the shift to libertarian values in advanced economies as well as specific questions about the unique case of Japan—where a shift toward gender egalitarianism (Inglehart and Norris, 2003) and individualism (Calista, 1984) has not occurred to the same extent as in other post-industrial countries. To what extent have values shifted in Japan in recent history and how has the economy influenced the nature and pace of social change? These are the questions that guide our analysis.

2. Change in values

The shift from materialist to post-materialist values has been studied extensively (Inglehart, 1971; Inglehart and Abramson, 1994; Inglehart and Norris, 2003; Taniguchi, 2006). Scholars have debated whether or not a shift to post-materialist values accompanies economic development in most countries (see, e.g., Clarke and Dutt, 1991; Inglehart and Abramson, 1999). Inglehart (1971) argues that as a society industrializes and people are able to meet their economic and physical

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security needs, individuals are able to look beyond their economic security to consider higher-order post-materialist values (values emphasizing intellectual and aesthetic needs, including personal freedom and quality of life). According to Inglehart and Baker (2000, p. 19), "Economic development is associated with shifts away from absolute norms and values toward values that are increasingly rational, tolerant, trusting, and participatory."

Other scholars have raised critiques of Inglehart and colleagues' research on value change. In particular, critiques have been raised of the measurement of post-materialist values (Clarke and Dutt, 1991), the measurement of economic period effects (Taniguchi, 2006), as well as the lack of attention paid to institutional context (Haller, 2002). Japan has been the focus of some of the critiques of Inglehart's work. Flanagan (1982) argued that value change in the case of Japan (through the mid-1970s) was not uni-dimensional but rather has occurred along two dimensions (from materialism to post-materialism and from authoritarianism to libertarianism). Both Inglehart (1982) and Calista (1984) caution that value change in Japan through the 1970s followed a different course than that observed in Western countries.

Inglehart and colleagues argue that values shift through processes of cohort replacement and in response to period effects. Cohort replacement is based on the idea that, to a large extent, an individual's values reflect the socio-economic conditions during their youth (Inglehart, 1971; Mannheim, 1952; Ryder, 1965). Consequently, there is some stability to values over an individual's life. Social change occurs as older cohort groups die and are replaced in the population by younger cohort groups. According to Inglehart and colleagues, values in a population become increasingly post-materialist as the country develops economically, largely because cohorts born and raised in relative economic deprivation at an earlier stage of development are replaced in the population by more recent cohorts born and raised in the relative affluence of a more advanced stage of economic development.

In addition to this gradual cohort replacement as a mechanism of social change in beliefs, period effects also motivate change. Beliefs in the population fluctuate over time in response to economic conditions and political and social events. For example, economic downturns are associated with a return to materialist value orientations (Abramson and Inglehart, 1995; Sakamoto, 2000) while political elections have been associated with short-term increases in post-materialism (Taniguchi, 2006). According to the scarcity hypothesis (Abramson and Inglehart, 1995), it is expected that during periods of prosperity there will be increased levels of post-materialism as individuals feel free to focus on higher-order needs, freedom, and autonomy but that during periods of scarcity, and the increased anxiety surrounding financial stability, levels of materialism will increase. Ultimately, social change in beliefs occurs through a combination of the more gradual process of cohort replacement and the within-cohort changes in response to key period effects. Later work by Inglehart and colleagues refined this model of social change by emphasizing the importance of cultural heritage in shaping a particular country's trajectory of change (Inglehart and Baker, 2000; Inglehart and Norris, 2003).

Other scholars have contributed to our understanding of social change through their analysis of the nature and pace of such change (Alwin et al., 1996, unpublished results, Braun et al., 1994; Lee et al., 2007, 2010, unpublished results-a,b; Mason and Lu, 1988; Scott et al., 1996). Using decomposition methods, these researchers have illuminated how the mechanisms of social change have varied over time (e.g., Mason and Lu, 1988) and cross-culturally (e.g., Lee et al., 2007, 2010; Scott et al., 1996). In particular, this literature has connected the pace of social change to key period effects and has also identified historical periods marked by great social change as well as those marked by relative stability.

Previous research in the US and Western Europe shows that in recent decades attitudes and beliefs have become increasingly more supportive of women's engagement in paid labor (see, e.g., Alwin and Scott, 1996; Alwin et al., 1996; Brewster and Padavic, 2000; Mason et al., 1976; Mason and Lu, 1988; Scott et al., 1993; Thornton et al., 1983; Thornton and Young-DeMarco, 2001). Values have also, more generally, become increasingly post-materialist (Abramson and Inglehart, 1995). With respect to beliefs about gender, changes in beliefs and attitudes have been experienced to a greater extent among women with higher levels of education and with labor-force experience (see, e.g., Thornton et al., 1983). While such changes have occurred for both women and men, changes for men have lagged behind those for women (e.g. Mason and Lu, 1988; Brewster and Padavic, 2000). These differences in the rates of change for groups defined by employment and educational status and by gender are attributed to the different "interests" of the different groups in gender equality (women, and working women in particular, have a greater stake in gender equality) as well as their different levels of "exposure" to gender-egalitarian ideology (those with a higher level of education are more likely to have been exposed to progressive beliefs about gender) (Bolzendahl and Myers, 2004).

3. Gender and the economy in Japan

Gender inequality is greater in Japan than in other advanced economies (see, e.g., Kinjo, 1995). This is evidenced in the work of Inglehart and Norris (2003) and Inglehart et al. (2002) which found higher levels of gender inequality and gendered beliefs in Japan vis-à-vis other post-industrial countries. Although approximately 50% of Japanese women (age 15 and up) are engaged in paid labor,[1] many working women are tracked into clerical jobs. A lack of support for working mothers and a lack of affordable child care for children under the age of three pushes many Japanese women out of the labor force after they have children (Yu, 2001). This combination of an advanced level of economic development and comparatively low levels of gender equality make Japan an ideal comparative case for analysis.

[1] Annual report of the Survey of the Labor Force. Statistical Survey Department, Statistics Bureau, Ministry of Internal Affairs and Communications.

A case study of social change in beliefs about gender in Japan provides a desirable comparison to studies of change in beliefs and attitudes in Western Europe and North America (see, e.g., Alwin and Scott, 1996; Brewster and Padavic, 2000; Mason and Lu, 1988). This research follows other recent research on social change in gender beliefs in Japan (Lee et al., 2010). Lee et al. (2010) found some divergences, from what would be expected based on research on Western countries, in the nature and pace of social change in beliefs about gender in Japan in the 1990s and early 21st century. This difference was attributable to the social-institutional context in which gender beliefs are formed in Japan, namely disincentives for women to work full-time and a high societal valuing of the housewife role. In the current paper, we set out to build on these findings and to examine more specifically how values have changed with fluctuating economic conditions from the 1970s to 2001 in Japan. This research also builds on earlier work examining social change in Japan through the 1970s, during Japan's record economic growth in the postwar period (see, e.g., Calista, 1984; Flanagan, 1982).

During the period of analysis for this paper, great social and economic changes were occurring in Japan. A new feminist movement coalesced in the early 1970s out of dissatisfaction with the treatment of women in Japanese society and even in leftwing political movements (Mackie, 1998). This movement has been successful in drawing greater public attention to gender inequality in Japanese society but still has considerable work to do in achieving its goals (Tanaka, 1995). In the 1980s, feminist activists were able to prevent legal changes that would have limited access to abortion, but their success with the Equal Employment Opportunity Law was more limited. The law that was passed in 1985 was toothless—making recommendations regarding workplace discrimination but without any explicit consequence for failure to comply (Matsui, 1990).[2] Still, women made gains in Japanese society and in the economy between 1970 and 1990 as evidenced by rising female labor force participation rates and the increased legitimacy of the women's movement (Tanaka, 1995). Despite these gains, however, a dual system of employment for men and women as well as continued support for a gendered division of labor in the population have hampered Japanese women's achievement of gender equality. Large firms in particular are unsupportive of working women have offer little flexibility in balancing work and family (Yu, 2005).[3] The economic collapse in the early 1990s proved to be an additional impediment in the progress towards gender equality.

Between 1986 and 1990, Japan experienced a bubble economy with commercial land prices increasing tenfold by 1989 and the Nikkei index of Japanese stocks tripling between 1985 and 1989 (Asher, 1996). The economic forecast at the time seemed to suggest unmitigated future growth and the Japanese business model was exported as a model of success to other countries. However, the bubble burst and stock prices began falling in 1990, with land prices following in 1991 (Tsuruta, 1999). By 1992, the extent of the financial crisis was fully realized when government measures could no longer buffer the worst effects of the economic downturn. According to Tanaka (1995: p. 351): "Women have been the first to be let go, while women graduating from college have had much greater difficulty than their male counterparts in finding jobs." The social movements as well as the booming economy that fueled women's opportunity and the economic downturn that followed likely influenced the nature and pace of social change in Japan from the mid-1970s to the 21st century.

4. Data and methods

We analyzed data from the Japan Institute of Life Insurance's Survey of Japanese People's Values in Life (1976–2001).[4] This national, random sample of adults age 16–69 includes measures of beliefs about the individual versus the collective and about tradition in five waves of data: 1976, 1985, 1991, 1996, and 2001 (total sample size = 7846).[5] These data allow for an analysis of social change in different types of beliefs during a period of economic turmoil in Japan.

In our analysis, we include measures of the respondents' beliefs regarding the importance of individual freedom and self-actualization in contrast to the importance of the survival of, and efficient production by, the group as a whole (e.g., the family, the workplace) as well as the importance of tradition. All of the measures of the beliefs used in this analysis are described in Table 1. In an exploratory factor analysis, two dimensions were found to underlie the individual indicators included in our analysis. The first dimension (which includes FEHOME, LINEAGE, and HARMONY) represents the tension between group and individual interests. The second dimension (which includes CUSTOM and CHANGEJOBS) more closely represents the tension between tradition and new ways of life. However, because we are interested in between-item differences, we analyze the individual indicators rather than the factor variables. We consider post-materialist and libertarian value orientations to be related and expect to see an overall shift toward increasingly post-materialist and libertarian values over the period of analysis. Because our measures focus more on personal freedom and individualism than on the material versus nonmaterial priorities of individuals, we will rely on Flanagan's (1982) terminology and refer to the shift from authoritarian to libertarian values. Beliefs about the appropriate roles for women and about the importance of lineage in choosing a spouse were asked in every wave of data collection, starting in 1976. All other measures were only asked in the 1985–2001 waves of data. For all of the variables included in the analysis, question wording did not change across survey years. In all of our analyses, we scale responses so that a high score reflects a libertarian value orientation. Disagreement with the belief that men should work

[2] The law was revised in 2007, however, addressing some of the weaknesses of the 1985 version.

[3] According to the 2006 Establishment and Enterprise Census, 48.3% of workers engaged in work at an establishment work in firms of 30 or more employees.

[4] The data for the secondary analysis, "Survey of Japanese People's Values in life, Japan Institute of Life Insurance" were provided by the Social Science Japan Data Archive, Information Center for Social Science Research on Japan, Institute of Social Science, the University of Tokyo.

[5] The sample was drawn through a stratified two-stage random sampling design. The response rates for each wave of the survey are as follows: 1976 (0.85), 1985 (0.72), 1991 (0.75), 1996 (0.72), 2001 (0.72).

Table 1
Indicators of beliefs in the Survey of Japanese People's Values in Life, 1976–2001.

Variable Name	Description
FEHOME	Women should stay at home after marriage
LINEAGE	I would like to choose my marital partner considering family lineage
HARMONY	When I do something, I want to respect harmony above my own opinion
CUSTOM	When I do something, I like to follow traditional practices
CHANGEJOBS	In order to take advantage of my abilities, I want to consider changing jobs

Note: All variables have been recoded so that a high score reflects libertarian values.

outside the home and women should specialize in domestic labor indicates a libertarian value orientation because the respondent is valuing a woman's independence and freedom over efficient household production.[6] Similarly, disagreement with beliefs about the importance of custom in shaping behavior and of the importance of harmony (as opposed to individual needs) in marital and work-related decisions indicates libertarian values. When a respondent prioritizes individual fulfillment and freedom over the needs of, or even the survival (in the case of questions of lineage) of, the group, this indicates a libertarian value orientation. Group harmony, loyalty to the firm, and valuing of custom and tradition are the hallmarks of traditional Japanese culture. Although these beliefs do not necessarily form a scale of post-materialist and libertarian values, we argue that rejection of these beliefs in favor of individual needs and desires represents a shift towards post-materialist and libertarian value orientations.

We first examined trends in the different measures of values between 1985 (or 1976 in the case of beliefs about gender and the importance of lineage in choosing a spouse) and 2001. We examined the overall changes in mean values and then the differences in mean values by arbitrarily chosen cohort groups to see if any of the change in values over time can plausibly be attributed to differences across birth cohort groups. We compared changes in some of the key measures of values alongside key economic indicators to look for evidence of a correspondence between changes in values and changes in the economy. Based on these descriptive analyses, we developed empirical predictions about the nature and pace of social change in Japan from 1985 (1976 in the case of beliefs about gender and the importance of lineage in choosing a spouse) to 2001 to be tested in a linear decomposition of social change in values.

Following the work of other scholars who have studied social change ("SC" or social change) through its decomposition into that part resulting from within-cohort change ("IC" for intra-cohort change) and that due to between-cohort differences ("CR" for cohort replacement) using a linear decomposition method (see, e.g., Alwin et al., unpublished results; Lee et al., 2007, 2010, unpublished results-a,b; Firebaugh, 1989), we decompose the change in measures of values observed in Japan between 1985 and 2001.[7] The slopes for survey year and for birth year in the linear regression model are used to compute the intra-cohort change and the cohort replacement which occurred between waves of data as follows:

$$\text{Intra-cohort change (IC)} = \beta_1(t_F - t_1) \text{ and}$$

$$\text{Cohort replacement (CR)} = \beta_2(C_F - C_1),$$

where β_1 is the slope for survey year, t_F is the final year of the period of analysis and t_1 is the first year, β_2 is the slope for birth year, C_F is the mean birth year in the final year of the period of analysis and C_1 is the mean birth year in the first year of the analysis. We first estimated this regression model without controls and then added controls for education (for both men and women) and for employment status (for women) to see if some of the intra-cohort change and cohort replacement is attributable to changes over time and across cohorts in levels of education and in the labor force participation rate for women. In these decompositions, education is measured as years of schooling completed and employment status for women is measured by a dummy variable: the respondent is employed full-time, compared to the reference category of being a student, working part-time, not being in the labor force or being unemployed.

The final stage of the decomposition analysis involved dividing the survey data into three periods of analysis: 1985–1991, 1991–1996, and 1996–2001. The 1985–1991 period spans the remarkable growth in the economy until its collapse in the early 1990s. The periods from 1991 to 1996 and then 1996 to 2001 span two periods of economic hard times in the aftermath of the asset price bubble's collapse. We also estimated the simulated social change that would have occurred in Japan from 1991 to 2001, were it not for the decreased level of intra-cohort change observed in these two periods.

[6] Although Braun et al. (1994) find that support for women's employment is based largely on economic necessity, Inglehart (1987) considers the women's movement to be a post-materialist issue and Inglehart and Abramson (1999) and Davis (2000) find evidence that post-materialists are more likely to support egalitarian gender beliefs. In this paper, we expect that disagreement with a gendered division of labor is an indicator of a post-materialist value orientation but will empirically examine this claim by analyzing the relationship between this belief and economic conditions.

[7] In the decomposition analysis, data from the 1985–2001 waves of data are included, in order to maintain consistency across measures.

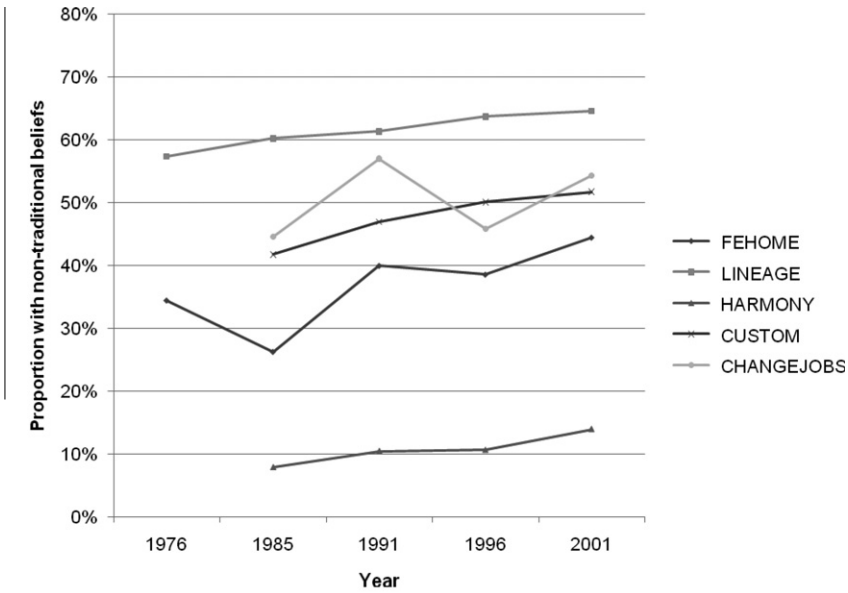
5. Results

5.1. Descriptive findings

Based on descriptive analyses of trends in mean values from 1976 to 2001, we see an overall pattern of gradually increasing acceptance of libertarian values in Japan (Fig. 1). In particular, respondents reported gradually increasing acceptance of libertarian beliefs about the importance of lineage in choosing a spouse, the importance of custom, and the importance of group harmony vis-à-vis individual opinion. Beliefs about gender and about the appropriateness of changing jobs to meet personal needs also became more libertarian over the period. However, these measures showed greater fluctuations from wave to wave of survey data, perhaps in response to changing economic conditions. Women reported more libertarian values about gender at the beginning and the end of the period of analysis, while men reported more libertarian values about the importance of lineage, harmony, and following custom at both time points.

When values are graphed by birth cohort instead of survey year (for survey years 1985–2001), we see a pattern of gradually increasing acceptance of libertarian values across birth cohorts (Fig. 2). Again, there are some notable differences with respect to the measure of gender beliefs and the measure of one's willingness to change jobs. We found a sharp increase in willingness to change jobs in the postwar cohorts compared to the prewar cohorts. With respect to beliefs about gender, in contrast, there is a steep increase in the acceptance of libertarian beliefs about gender between the birth cohort born before 1930 and those born 1941–1950 which levels out in subsequent birth cohorts. Overall, this graph suggests there are important cohort differences in beliefs. Based on this descriptive analysis, it seems that some of the increasing acceptance of libertarian beliefs observed in Fig. 1 is likely attributable to cohort replacement.

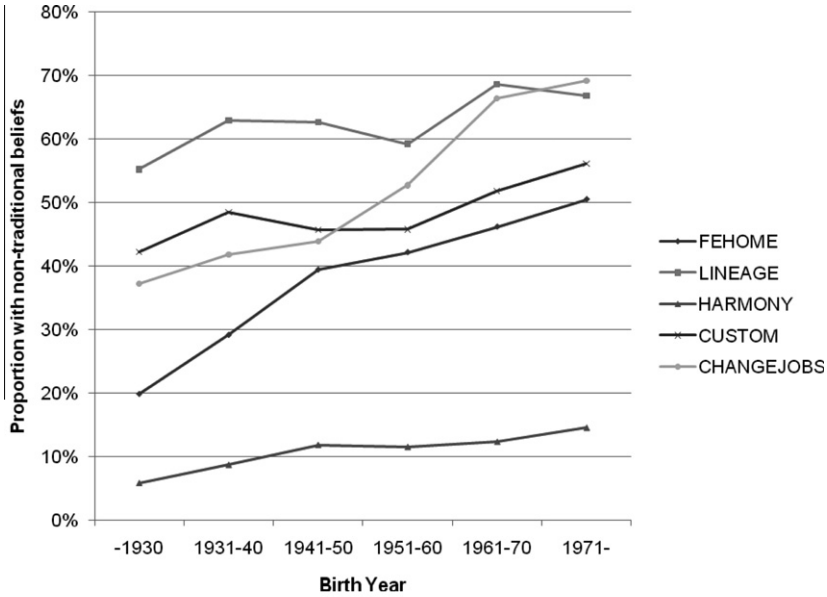
To gain some insight into the possible relationship between economic period effects and changing values, we also graphed the measures of beliefs that showed the greatest fluctuation over time (beliefs about gender and willingness to change jobs) with a measure of economic conditions (men's unemployment rate). As would be expected, when economic conditions are poor (e.g., men's unemployment rate is high), respondents indicated less willingness to change jobs, compared to periods of economic strength (Fig. 4). This relationship holds through 1996; however, between 1996 and 2001, willingness to change jobs increased despite increasing levels of male unemployment. Similarly, beliefs about gender also seem to mirror changes in the economy through 1996 (Fig. 3). As male unemployment increased in the late 1970s and into the mid-1980s, beliefs about gender became more authoritarian (or focused on the best interest of the family over the individual freedom of women). However, when unemployment leveled out and even fell in the mid-1980s and early 1990s, beliefs



Note: The following table provides the sample size for each data point in the graph.

	1976	1985	1991	1996	2001
FEHOME	931	1412	1837	1787	1788
LINEAGE	929	1384	1817	1776	1769
HARMONY		1406	1842	1784	1783
CUSTOM		1393	1816	1766	1779
CHANGEJOBS		1316	1760	1711	1727

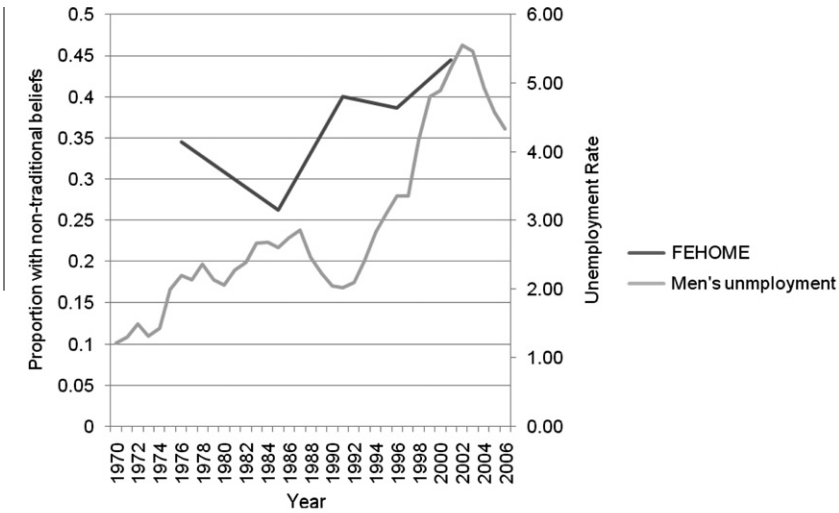
Fig. 1. Percentage of responses indicating individualistic beliefs by survey year. 1976–2001 Survey of Japanese People's Values in life.



Note: The following table provides the sample size for each data point in the graph.

	-1930	1931-40	1941-50	1951-60	1961-70	1971-
FEHOME	709	1419	1577	1368	1065	685
LINEAGE	695	1400	1560	1353	1056	681
HARMONY	698	1397	1575	1368	1076	700
CUSTOM	689	1400	1560	1347	1062	695
CHANGEJOBS	633	1304	1529	1325	1050	672

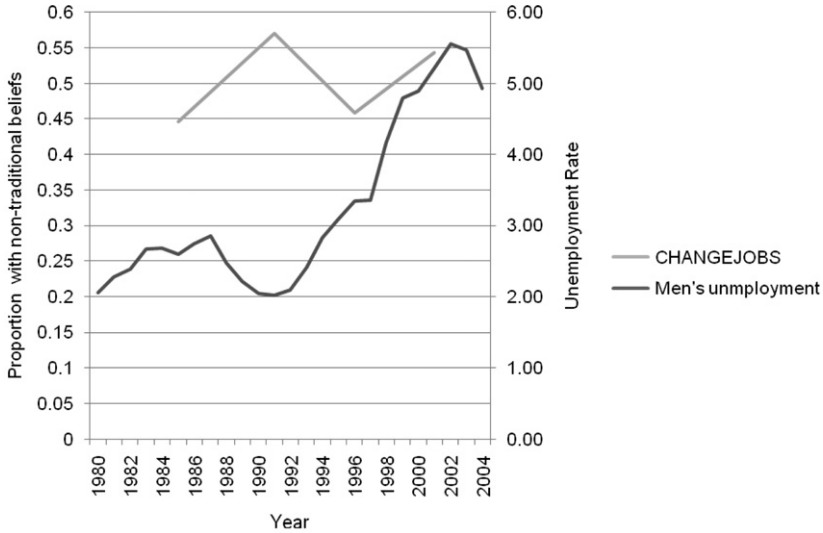
Fig. 2. Percentage of responses indicating individualistic beliefs by birth cohort group. 1985–2001 Survey of Japanese People's Values in life.



Note: Men's unemployment rate indicates the average seasonally adjusted unemployment rate.
Source: Statistics Bureau, Japanese Ministry of Internal Affairs and Communications.

Fig. 3. Trends in beliefs about gender (for men and women in the 1976–2001 Survey of Japanese People's Values in Life; $N = 7755$) and in the unemployment rate for men in Japan.

about gender became less authoritarian. With the burst of the bubble in the early 1990s, male unemployment once again rose and beliefs about gender became slightly more authoritarian until 1996. Although male unemployment continued to rise between 1996 and 2001, beliefs about gender became increasingly libertarian, counter to previous trends. These graphs



Note: Men's unemployment rate indicates the average seasonally adjusted unemployment rate.
Source: Statistics Bureau, Japanese Ministry of Internal Affairs and Communications.

Fig. 4. Trends in willingness to change jobs (for men and women in the 1985–2001 Survey of Japanese People's Values in Life; $N = 6514$) and in the unemployment rate for men in Japan.

suggest that economic period effects may explain part of the trends in changing values for men and women in Japan, but not the whole story. In order to understand both the nature and pace of social change in beliefs, we must decompose the change into that attributable to intra-cohort change and that attributable to cohort replacement.

5.2. Empirical predictions

Based on previous research and the descriptive analyses, several empirical predictions were made for the decomposition analysis. We expected that cohort replacement would contribute to the gradual increase in libertarian values in the analysis, consistent with previous research which documented declines in traditional values in Japan across cohorts born 1923–1972 (Atoh, 1997). We expected that social change in the measures of beliefs about gender and willingness to change jobs, in particular, was motivated by intra-cohort change as well as cohort replacement. The economic downturn experienced during the early 1990s was expected to reduce willingness to change jobs and to inspire a return to more traditional gender roles. We also expected that intra-cohort change played an important role during the earlier periods of analysis as Japan was rapidly modernizing and growing but that the pace of social change slowed in more recent years as economic hard times inspired some return to authoritarian or materialist values, or at least a slowdown in the adoption of libertarian or post-materialist values, and as social change has come to be motivated more by the gradual process of cohort replacement and less by intra-cohort change. This is consistent with previous research which found a slowdown in the adoption of post-materialist values during the oil crisis in the 1970s (Sakamoto, 2000).

5.3. Decomposition analysis

Overall, for the period from 1985–2001, there is statistically significant social change in all measures, in the direction of increasing acceptance of libertarian values (see Table 2). The social change with the greatest magnitude observed during this period occurred with respect to beliefs about gender. Beliefs about gender changed at least twice as much as beliefs about the importance of lineage in choosing a spouse, about the importance of maintaining group harmony, and about the importance of following custom. Over the entire period of observation, men experienced more social change in beliefs about gender, as well as in their willingness to change jobs, than women did. This difference is not, however, statistically significant at the 5% level. For men, about 65% of the change in beliefs about gender occurred through intra-cohort change whereas for women, about 52% of the change was attributable to intra-cohort change. Beliefs about the importance of lineage, of protecting harmony, and of following custom also changed primarily through intra-cohort change for women; beliefs about changing jobs changed primarily through cohort replacement. This finding about willingness to change jobs is surprising and will be explored further in the period analysis. For men, on the other hand, beliefs about lineage, harmony, and changing jobs all changed primarily through the slower process of cohort replacement.

When controls for education are added to the regression models, the size of the coefficients change (largely because of cohort differences in educational attainment that underlie some of the cohort replacement). Generally speaking, the contribution of cohort replacement to social change, relative to intra-cohort change, decreases once education is controlled. The

Table 2
Decomposition of social change, full-time period (1985–2001), $N = 6824$.

	No controls								
	All respondents			Men			Women		
	SC	IC	CR	SC	IC	CR	SC	IC	CR
FEHOME	.499***	.262***	.195***	.535***	.331***	.182***	.476***	.224***	.210***
LINEAGE	.174***	.097**	.081***	.129**	.038	.103***	.207***	.144**	.070***
HARMONY	.229***	.128***	.098***	.196***	.077	.113***	.254***	.160***	.093***
CUSTOM	.220***	.161***	.061***	.236***	.157***	.071***	.203***	.160***	.058***
CHANGEJOBS	.259***	−.048	.219***	.321***	.012	.218***	.203***	−.096*	.221***
	Controls for education Men			Controls for education Women			Controls for education and employment Women		
	SC	IC	CR	SC	IC	CR	SC	IC	CR
FEHOME	.535***	.344***	.153***	.476***	.238***	.157***	.476***	.243***	.156***
LINEAGE	.129**	.023	.114***	.207***	.124**	.086***	.207***	.126**	.085***
HARMONY	.196***	.086*	.099***	.254***	.170***	.092***	.254***	.171***	.092***
CUSTOM	.236***	.163***	.056***	.203***	.157***	.052***	.203***	.156***	.051***
CHANGEJOBS	.321***	.017	.199***	.203***	−.082	.198***	.203***	−.087	.198***

* $p < .10$.
** $p < .01$.
*** $p < .001$.

additional control for employment in the case of women makes little difference in the results. We also conducted tests of alternative measures of employment for women (i.e., occupation and firm size) but the results are largely the same (results not shown).

A decomposition analysis looking across the entire period of 1985–2001 misses some of the important period differences within this large time span. To capture these changes in Japanese society and economy, a decomposition of social change in values was conducted for three periods: 1985–1991, 1991–1996, and 1996–2001. Controls for education (and for employment for women) are included in these analyses. When the decomposition is divided into three periods, we see that most of the social change in values occurred in the first and last periods, with relatively little social change occurring between 1991 and 1996 (see Table 3). In fact, the only statistically significant social change during this period for men was with respect to willingness to change jobs and for women was with respect to values regarding following custom and willingness to change jobs. For both women and men, social change resulted in a decreased willingness to change jobs during the period immediately following the collapse of the bubble economy. The social change in willingness to change jobs occurred almost entirely through intra-cohort change. Although differences across cohorts resulted in a small increase in willingness to change jobs, all of the decrease in willingness to change jobs occurred through intra-cohort change, most likely in response to economic period effects. This is consistent with the hypothesis that willingness to change jobs would be particularly responsive to period effects. The decomposition analysis of social change over the entire period from 1985 to 2001 seemed to refute the hypothesis that social change in willingness to change jobs occurred primarily through intra-cohort change. However, once the period of analysis was divided into three periods closely linked to economic cycles, it became clear that the seeming absence of much intra-cohort change was simply an artifact of counterbalancing positive intra-cohort change in the first period and negative intra-cohort change in the second period. Most other values remained largely unchanged during this period from 1991 to 1996. The economic collapse seems to be associated with a stall in the transformation of values regarding Japanese custom and tradition as well as gender.[8] The association between egalitarian gender beliefs and economic good times supports our claim that changes in beliefs about gender indicate libertarian or post-materialist values rather than an acknowledgment of the economic necessity of two incomes.[9]

This period of no change from 1991 to 1996 is a contrast to both the preceding (1985–1991) and following (1996–2001) periods. During the first period in particular, considerable social change occurred for both men and women with respect to increasingly libertarian beliefs about gender, the importance of following custom, and in willingness to change jobs. For women, statistically significant social change in beliefs about the importance of lineage in choosing a spouse also occurred in

[8] The robustness of these findings was tested through an alternative specification of the periods of analysis. In an alternative analysis in which change from 1985 to 1991 was compared to change from 1991 to 2001 (results not shown), the argument that the adoption of post-materialist values slowed during the economic hard times starting in the early 1990s was tested. Although the period from 1991 to 2001 is nearly twice the length of time as the period from 1985 to 1991, the rate of social change in most measures remained constant or even declined in this longer second period (with the exception of social change in HARMONY which did increase for women and men in the second period and also LINEAGE for men). Similarly, we would have expected to see double the amount of intra-cohort change in the second longer period but instead observed the same amount or a decline in the amount of intra-cohort change observed in the first period (with the exception of IC in CUSTOM for women).

[9] This claim is further supported by additional analyses (available from the author upon request) showing that when log household income is added to models predicting beliefs about gender, it is positively associated with egalitarian gender beliefs. If gender beliefs were instead acting as an indicator of a materialist orientation, we would expect these variables to be negatively related.

Table 3
Decomposition of social change by time period (1985–1991, 1991–1996, 1996–2001), $N = 6824$.

	Men			Women		
	SC	IC	CR	SC	IC	CR
1985–1991						
FEHOME	.289***	.226***	.069***	.382***	.295***	.070***
LINEAGE	.010	−.057	.059***	.127**	.093*	.044***
HARMONY	.060	.018	.046***	.044	.006	.042***
CUSTOM	.130**	.079	.043***	.123**	.100*	.023*
CHANGEJOBS	.340***	.260***	.084***	.273***	.188***	.078***
1991–1996						
FEHOME	.076	.079	.008***	−.052	−.058	.002***
LINEAGE	.058	.038	.007***	.036	.017	.001***
HARMONY	.011	.007	.006***	.070	.065	.002***
CUSTOM	.045	.054	.003**	.092*	.087*	.001**
CHANGEJOBS	−.221***	−.230***	.012***	−.211***	−.207***	.003***
1996–2001						
FEHOME	.170***	.061	.089*	.146**	.059	.081***
LINEAGE	.061	.024	.045***	.044	.023	.040***
HARMONY	.125**	.061	.051***	.140***	.096*	.047***
CUSTOM	.061	.035	.018	−.012	−.044	.030**
CHANGEJOBS	.202***	.084	.108***	.140**	.009	.114***

Note: Decompositions for men include controls for education and for women include controls for education and full-time employment status.
* $p < .10$.
** $p < .01$.
*** $p < .001$.

this first period. Although intra-cohort change contributed significantly only to social change in beliefs about gender and in willingness to change jobs for men and women (and to social change in beliefs about the importance of following custom for women), when it was statistically significant, it motivated most of the social change that occurred for those beliefs. About 76% of the social change in beliefs about gender for women and men during the earliest period was attributable to intra-cohort change, as well as 70–76% of the change in willingness to change jobs for women and men, respectively. This period of great social change in beliefs about gender and about tradition and custom coincided with a period of economic prosperity.

During the third period of the analysis (1996–2001), Japan was still facing economic hard times and had moved further away from the period of its remarkable economic growth. Social change began to pick up again, albeit at a lower level than in the pre-collapse days of the late 1980s. Men and women experienced statistically significant social change in beliefs about gender, the importance of harmony, and in willingness to change jobs. The level of social change in values was lower in the third period with its weak economy than in the first, economically robust period and was motivated more by cohort replacement than by intra-cohort change. While the bulk of social change in beliefs about gender and in willingness to change jobs was motivated by intra-cohort change in the first period, no statistically significant intra-cohort change was found for these two measures during the third period. Instead, change in most measures occurred through the slower process of cohort replacement during the last time period. The only exception to the finding that cohort replacement dominated in the last time period is that women experienced statistically significant intra-cohort change with respect to values regarding the importance of harmony.[10]

There is evidence of a slowdown in the pace of social change in values in Japan, starting in the 1990s. Although there is evidence of change in the late 1990s and early 21st century, the pace of change is still slower than in the 1980s. This slowdown does not seem to be attributable to a ceiling effect since none of the measures have a mean that surpasses four on a scale of five (with five indicating strong agreement with libertarian values). Also, while a ceiling effects explanation might explain an overall, general slowdown, it does not explain the stalled social change we observed in the 1990s, followed by a return to the adoption of libertarian values in the late 1990s and early 21st century. Instead, we argue that economic period effects are an important part of the story of slowing social change in values in Japan.

To examine what social change would have looked like in Japan, were it to have maintained the same pace uninterrupted by the period effects restraining change in the 1990s, we conducted a simulated decomposition of social change. We first used the levels of cohort replacement observed in periods two and three (reported in Table 3), and substituted the levels of intra-cohort change from the first period for those actually observed for the second and third period in conducting this simulation (separately for men and women). This allows us to compare the rate of social change in values observed in the second and third periods with the rate we would have seen if intra-cohort change (driven primarily by period effects) had continued at the pace observed in the late 1980s in the 1990s. What we found is that, for most measures, social change in the direction of increasingly libertarian values would have been much greater had it not been for the dip in intra-cohort change during the second two periods. In Fig. 5 we show the results of the simulation for the two indicators graphed in Figs. 3

[10] These periodized results were also tested controlling for the respondent's occupation and in a sub-sample of respondents employed in firms with fewer than 30 employees. The substantive findings do not change with these alternative measures of employment.

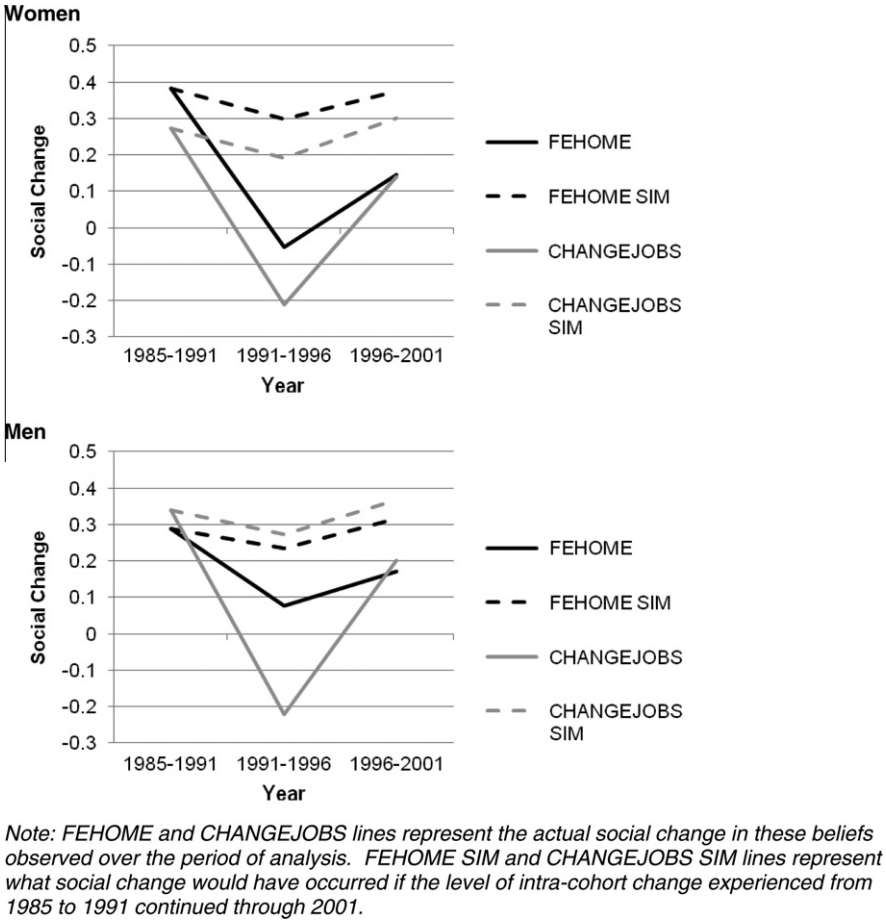


Fig. 5. Simulated social change in beliefs in Japan from 1991 to 1996 and 1996 to 2001, with intra-cohort change fixed at the level of change experienced from 1985 to 1991.

and 4—FEHOME and CHANGEJOBS. With respect to beliefs about gender, men would have experienced 2.2 times more social change from 1991 to 2001 in this belief had intra-cohort change continued at the rate observed from 1985 to 1991, and women would have experienced 7.2 times more change. This is in comparison to a simulation in which the rate of cohort replacement is artificially kept constant at the 1985–1991 level and the observed levels of ICC in periods 2 and 3 are maintained. In this simulation, men would have experienced approximately the same amount of change as that actually observed between 1991 and 2001 and women would have experienced only 1.5 times more change during periods 2 and 3. This simulation supports the argument that most of the reduction in the rate of social change in the second two periods of analysis is attributable to changes in the rate of intra-cohort change. Period effects, and not reductions in cohort difference or in the rate of cohort replacement, explain most of this slowdown in social change.

6. Discussion

We set out to uncover the nature and pace of social change in Japan from 1976 to 2001, with particular attention to the interplay between values and the economic crisis of the early 1990s. Given Japan's persisting gender inequality in education and work, we were interested in explaining how much change occurred in beliefs about gender, vis-à-vis other beliefs during this period, as well as investigating the pace of change in periods of both economic prosperity and hard times. Our initial descriptive analyses suggested that certain beliefs were more responsive to period effects than others, with willingness to change jobs and beliefs about gender roles fluctuating most with changing economic conditions. Overall, in the decomposition analyses, we found evidence of increasingly libertarian and post-materialist beliefs in the population over the period of analysis (consistent with Inglehart's theory of value change), with beliefs about gender and willingness to change jobs changing the most for men and with beliefs about gender and beliefs about maintaining harmony changing the most for women. Overall, social change for the entire period occurred through both cohort replacement and intra-cohort change.

When the analysis was divided into three periods that correspond to important economic periods, a more nuanced understanding of social change in Japan emerged. In particular, trends in values appear connected to changes in economic conditions. Overall, the greatest social change in values occurred during the economic good times of the late 1980s. Following the burst of the asset price bubble, social change stalled only to resume at a much slower pace in the late 1990s.

Based on a simulation analysis, we found that, were it not for the slowdown in intra-cohort change in the 1990s, values in Japan would have been considerably more libertarian by the year 2001, all else equal. The slower process of cohort turnover continued to contribute to the adoption of individualistic values as prewar birth cohorts raised with relatively more materialist and authoritarian value orientations emphasizing group success and survival over individual needs and freedom were replaced in the population by more recent cohorts born in the postwar period. Period effects (we suspect largely attributable to the economic downturn), however, led to a reduction in the rate of social change (and even a stalling of social change) in the last two periods.

This research contributes to both our understanding of recent social change in Japan and also the relationship between the economy and value orientations. Recent research on Japan found that while measures of gender ideology became increasingly egalitarian between 1994 and 2002, measures of the importance of women working became less egalitarian (Lee et al., 2010). It was speculated that perhaps this divergence in trends was attributable to conflict between beliefs about gender egalitarianism and increasing individualism. The current paper provides support for the claim that Japanese respondents reported increasingly individualistic, libertarian values from 1976 to 2001. In particular, increasing rejection of the valuing of group harmony over individual opinion, of following traditional practices, and of prioritizing group interests through the selection of a spouse based on family lineage all indicate a trend toward individualism and libertarianism in Japan. Also, in the analysis of social change during the period from 1985 to 2001, we found additional support for the argument made by Lee et al., 2010 that in recent history, men's changing gender ideology was more closely connected to economic period effects than was women's. However, in the current paper, this finding did not apply equally to all time periods. Both men and women only experienced statistically significant intra-cohort change in the first period of analysis (1985–1991), and during this first period, intra-cohort change represented a larger proportion of the total social change in beliefs about gender for men than for women.

As for the larger question of the relationship between the economy and libertarian and post-materialist values, this research suggests that they are intimately linked. In our analysis of social change in values, we did not observe a linear trend of increasingly libertarian values, but rather found that in economic hard times, advancement towards libertarianism stalled and even reversed in the case of willingness to change jobs. Future research should test these findings with the indicators of a post-materialist value orientation identified by Inglehart (see, e.g., Inglehart and Abramson, 1999). An analysis of the magnitude of the effect of changing economic conditions compared to other period effects would also be desirable. This analysis was designed to examine the relationship between poor economic conditions and social change in values in a post-industrial country. Future research should examine the connection between the economy and values in other post-industrial countries, as well as in countries at different stages of economic development.

Japan has experienced considerable social change with respect to gender roles, the family, and values regarding tradition and custom since WWII. At the same time, compared to other countries with similarly advanced economies, Japan lags behind its peers with respect to gender equality and value change (Inglehart and Norris, 2003; Inglehart et al., 2002). We argue that at least some of this lag in value change is attributable to poor economic conditions that plagued Japan during the 1990s. Although we cannot say unequivocally what would have happened if the economic crisis had not intervened, we speculate that greater levels of social change in beliefs about gender and the importance of custom and tradition would have occurred.

National economic conditions are not the only explanation of social change in Japanese values, however. We discussed the feminist movement that developed in the 1970s; this certainly played a role in the adoption of libertarian values, especially beliefs about gender. We suspect that global interaction with other post-industrial economies that have made progress towards gender equality, the need for two incomes, labor market deregulation, and socio-legal changes which have made it somewhat easier for women to work outside the home are an important part of this story. Deregulation of the labor market in the mid-1990s and legislation promoting gender equity in the workplace in the 1980s are particularly important to consider in understanding changing values. It is important to note that these institutional changes in the labor market primarily impact employees of large firms and cannot explain the slowdown in the adoption of libertarian values following the burst of the asset price bubble. These institutional changes would be expected to increase acceptance of changing jobs and increase acceptance of women working outside the home but, in fact, we saw a decline in social change in values in the 1990s compared to the economic good times of the 1980s. Certainly these changes in the labor market played a role, particularly in motivating the social change observed between 1996 and 2001, but the role of the economy cannot be denied in explaining social change in values in the 1990s and into the 21st century in Japan. Although we cannot adjudicate between the significance of the economy vis-à-vis socio-legal changes, we think this is an important direction for future research.

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